

Mathematical Formalism of Cayley–Dickson Zero-Divisor Structure

open_gororoba Project

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1 Scope and Validation Status

This document describes the mathematical formalism underlying the `open_gororoba` codebase. It separates:

- standard definitions (Cayley–Dickson construction, associator),
- computationally validated facts (verified by Rust unit tests in `crates/algebra_core/`), and
- speculative proposals that are clearly marked as such.

Treat any statement as non-authoritative unless backed by a first-party citation and a reproducible test under `crates/*/src/` or the claims evidence matrix (`registry/claims.toml`).

2 Cayley–Dickson Construction (Standard)

2.1 Recursive Definition

Let A_n be the algebra obtained after n iterations of the Cayley–Dickson process, with $\dim(A_n) = 2^n$. The standard doubling is:

$$A_{n+1} = A_n \oplus A_n, \quad (a, b)(c, d) = (ac - \bar{d}b, da + b\bar{c}) \quad (1)$$

3 Trilinear Order-Sensitivity Operators

3.1 Associator (Standard)

For octonions and beyond, associativity fails. The standard associator is:

$$[x, y, z] = (xy)z - x(yz) \quad (2)$$

3.2 The m_3 Operator (Validated Properties)

The repo computes a specific trilinear operator m_3 (defined in `crates/algebra_core/src/m3_transfer.rs`). On distinct octonion basis triples (e_i, e_j, e_k) with $i, j, k \in \{1, \dots, 7\}$ all distinct, the following properties are validated:

- exactly 42 scalar outputs and 168 pure-imaginary outputs, and
- the scalar cases correspond to the Fano-plane lines (with parity sign flips).

These are validated by `crates/algebra_core/src/m3_transfer.rs` test module. Establishing an A_∞ structure would require additional, separately scoped Stasheff-identity checks.

4 Sedenion Zero Divisors

4.1 The 42 Assessors

In the de Marraais construction, there is a set of 42 primitive cross-assessors that generate a diagonal-form family of 2-blades. This repo replicates:

- 42 primitive assessors, and
- a partition into 7 box-kites (6 assessors each)

under the repo's fixed 16D Cayley–Dickson convention, validated by `crates/algebra_core/src/boxkites.rs` test module.

4.2 Box-Kite Structure

Box-kite is a specific combinatorial object: the complete tripartite graph $K_{2,2,2}$. At $\dim = 16$, all 7 connected components of the cross-assessor zero-product graph are isomorphic to $K_{2,2,2}$.

5 Results Established Since January 2026

The following items were listed as “speculative directions” in the original January 2026 draft. Their status has been resolved:

Motif census (dim=32/64/128/256) COMPLETED (Sprint 6). Exact scaling laws verified across 5 doublings. No octahedra beyond $\dim = 16$; structure completely reorganizes at each doubling. See `crates/algebra_core/src/boxkites.rs`, claims C-126 through C-130.

Reggiani geometric identifications PARTIALLY COMPLETED (Sprint 9). 8 monograph theses (A–H) verified, including lattice codebook filtration and parity constraints. Full geometric mapping (PSL(2,7) action) remains open.

E8 root-system mapping REFUTED (Sprint 8). Zero out of 336 pairwise lattice-point differences have norm-squared 2. The minimum lattice separation for ZD-adjacent pairs is $|d|^2 = 4$. See claim C-455.

6 Open Questions

- **Twist-delta pair correspondence** (C-466): relationship between de Marraais twist pairs and lattice codebook structure.
- **PSL(2,7) geometric mapping**: explicit construction of the projective special linear group action on box-kite coordinates.
- **Higher-dimension emanation**: extending L1–L18 emanation layers beyond $\dim = 16$ to $\dim = 32$ and beyond.